EXPERIMENT-9

Compare Linear Regression and Polynomial Regression

Code:

```
import pandas as pd
from google.colab import files
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures

print("="*70)
print("     LINEAR VS POLYNOMIAL REGRESSION")
print("="*70)

uploaded=files.upload()

data=pd.read_csv("Salary_Data.csv")

print("\nDataset")
print(data)

X=data[['YearsExperience']]
y=data['Salary']

linear=LinearRegression()

linear.fit(X,y)

poly=PolynomialFeatures(degree=2)
```

```
X_poly=poly.fit_transform(X)

poly_model=LinearRegression()

poly_model.fit(X_poly,y)

sample=[[6]]

linear_salary=linear.predict(sample)

poly_salary=poly_model.predict(poly.transform(sample))

print("\nLinear Regression Prediction")
print(round(linear_salary[0],2))

print("\nPolynomial Regression Prediction")
print(round(poly_salary[0],2))

print("\nDifference")
print(round(abs(linear_salary[0]-poly_salary[0]),2))

print("\nComparison Completed Successfully")
```

Sample Input:

```
...
======================================================================
     LINEAR VS POLYNOMIAL REGRESSION
======================================================================
Choose Files  Salary_Data.csv
Salary_Data.csv(text/csv) - 454 bytes, last modified: 8/5/2026 - 100% done
Saving Salary_Data.csv to Salary_Data (1).csv
```

Sample Output:

```
Dataset
    YearsExperience    Salary
0               1.1   39343.0
1               1.3   46205.0
2               1.5   37731.0
3               2.0   43525.0
4               2.2   39891.0
5               2.9   56642.0
6               3.0   60150.0
7               3.2   54445.0
8               3.2   64445.0
9               3.7   57189.0
10              3.9   63218.0
11              4.0   55794.0
12              4.0   56957.0
13              4.1   57081.0
14              4.5   61111.0
15              4.9   67938.0
16              5.1   66029.0
17              5.3   83088.0
18              5.9   81363.0
19              6.0   93940.0
20              6.8   91738.0
21              7.1   98273.0
22              7.9  101302.0
23              8.2  113812.0
24              8.7  109431.0
25              9.0  105582.0
26              9.5  116969.0
27              9.6  112635.0
28             10.3  122391.0
29             10.5  121872.0

Linear Regression Prediction
82491.97

Polynomial Regression Prediction
82360.77

Difference
131.21

Comparison Completed Successfully
```